Lecture 5

L^p spaces, convolution and smoothing

Objectives. *A wavefunction in L^2 is specified up to changes on a null set. We prove that limits in integral norm remain in L^p. We then approximate these classes by smooth functions whose supports stay inside the given open set. These results supply the function spaces used in Fourier analysis and quantum mechanics.*

By the end of this lecture you will be able to:

- Apply Hölder's and Minkowski's inequalities, including their endpoints.
- Prove completeness and extract an almost-everywhere convergent subsequence.
- Justify convergence of an approximate identity in a stated norm.
- Approximate an $L^p(\Omega)$ function by compactly supported smooth functions when $p < \infty$.

5.1 Classes and integral norms

Let $E \subset \mathbb{R}^d$ be measurable. For $1 \le p < \infty$, the space $L^p(E)$ consists of measurable complex functions with finite norm

$$\|f\|_p = \left(\int_E |f|^p \right)^{1/p},$$

where functions equal a.e. represent the same element. The space $L^\infty(E)$ consists of the essentially bounded classes. Its norm is the *essential supremum*, defined by

$$\|f\|_\infty = \inf\{M \ge 0 : |f(x)| \le M \text{ for a.e. } x \in E\}.$$

The inequality $|f| \le \|f\|_\infty$ holds outside a null set: take the union of the null sets where $|f| > \|f\|_\infty + 1/j$, for $j \in \mathbb{N}$. Thus these definitions ignore exceptional values. For example, changing $f(0)$ from zero to a million does not change any of its L^p norms on an interval. Positive homogeneity is immediate. The norm vanishes exactly on the zero class. For finite p, use the criterion in § 3.3: a nonnegative measurable function has integral zero exactly when it vanishes a.e. For $p = \infty$, use the essential bound.

The *conjugate exponent* p' is defined by $1/p + 1/p' = 1$, with $1/\infty = 0$. This prime denotes an exponent, not a derivative.

Theorem 5.1 (Hölder's inequality). *For $1 \le p \le \infty$, $f \in L^p(E)$ and $g \in L^{p'}(E)$,*

$$\|fg\|_1 \le \|f\|_p \, \|g\|_{p'}.$$

Proof. For $1 < p < \infty$, maximising $ab - a^p/p$ over $a \ge 0$ gives $a^{p-1} = b$ and the maximum $b^{p'}/p'$. Hence $ab \le a^p/p + b^{p'}/p'$ for $a, b \ge 0$. If both norms are nonzero, apply this inequality to $a = |f| / \|f\|_p$ and $b = |g| / \|g\|_{p'}$ and integrate. The resulting bound is $1/p + 1/p' = 1$. If

either norm is zero, its function is zero a.e. For the endpoint $p = 1$, integrate $|fg| \le |f| \, \|g\|_\infty$. Interchanging the functions proves the endpoint $p = \infty$. ∎

For example, on $(0, 1)$ the function $f(x) = x$ satisfies $\|f\|_p = (p+1)^{-1/p}$ for finite p, and $\|f\|_\infty = 1$. Hölder with $g = 1$ gives $1/2 \le (p+1)^{-1/p}$.

Theorem 5.2 (Minkowski's inequality). *For $1 \le p \le \infty$ and $f, g \in L^p(E)$,*

$$\|f + g\|_p \le \|f\|_p + \|g\|_p .$$

Thus $L^p(E)$ is a normed vector space.

Proof. For finite p, the bound $|f+g|^p \le 2^p(|f|^p + |g|^p)$ first proves integrability. For $1 < p < \infty$, apply Hölder to each term in

$$\int_E |f+g|^p \le \int_E (|f| + |g|) \, |f+g|^{p-1} \le (\|f\|_p + \|g\|_p) \, \|f+g\|_p^{p-1} .$$

Here $(p-1)p' = p$. Divide unless the last norm is zero. For $p = 1$, integrate the pointwise triangle inequality. For $p = \infty$, apply it off the union of the two exceptional null sets. ∎

5.2 Completeness and almost-everywhere limits

Recall that a *Banach space* is a complete normed vector space. Completeness means that every Cauchy sequence converges in the norm.

Theorem 5.3 (Riesz–Fischer). *For $1 \le p \le \infty$, the space $L^p(E)$ is complete. Every sequence converging in $L^p(E)$ has a subsequence converging a.e. to a representative of its norm limit.*

Proof. First let $p < \infty$ and let (f_n) be Cauchy. Choose increasing indices n_j with $\|f_{n_{j+1}} - f_{n_j}\|_p \le 2^{-j}$. Choose measurable representatives, and put

$$h_N = \sum_{j=1}^{N} |f_{n_{j+1}} - f_{n_j}| , \qquad h = \sum_{j=1}^{\infty} |f_{n_{j+1}} - f_{n_j}| .$$

Minkowski gives $\|h_N\|_p \le 1$. Monotone convergence, Theorem 3.2, applied to h_N^p gives $\int_E h^p \le 1$. Thus h is finite a.e. The telescoping series

$$f = f_{n_1} + \sum_{j=1}^{\infty} (f_{n_{j+1}} - f_{n_j})$$

converges absolutely there. Set its value to zero on the exceptional null set. It is measurable and belongs to L^p, since $|f| \le |f_{n_1}| + h$ a.e. The same argument on the tail gives

$$\|f - f_{n_J}\|_p \le \sum_{j=J}^{\infty} 2^{-j} = 2^{1-J} .$$

The subsequence therefore converges in norm and a.e. For any n, the triangle inequality bounds $\|f_n - f\|_p$ by $\|f_n - f_{n_J}\|_p + \|f_{n_J} - f\|_p$. The Cauchy property and then large J prove convergence of the full sequence.

For $p = \infty$, discard the countable union of the exceptional sets in $|f_n - f_m| \le \|f_n - f_m\|_\infty$ and $|f_n| \le \|f_n\|_\infty$, for all $n, m \in \mathbb{N}$. On the complement, (f_n) is uniformly Cauchy and

uniformly bounded. Its uniform limit f is measurable and bounded there. Define $f = 0$ on the null set. Passing $m \to \infty$ in the pointwise bounds proves $\|f_n - f\|_\infty \to 0$.

Finally, apply these constructions to any norm-convergent sequence. For finite p they provide an a.e.-convergent subsequence whose norm limit equals the prescribed limit, by uniqueness in a normed space. For $p = \infty$ the whole sequence converges a.e. after the countable null set is removed. ∎

The typewriter sequence of § 3.5 shows why only a subsequence can be promised when $p < \infty$. Its terms converge in every finite $L^p(0,1)$, but at no point do their values converge.

5.3 Continuous approximation and translation

For an open set $\Omega \subset \mathbb{R}^d$, the space $C_c(\Omega)$ consists of continuous functions whose *support*, the closure of their nonzero set, is a compact subset of Ω.

Theorem 5.4 (M6: continuous approximation · *cited*)**.** *For every open $\Omega \subset \mathbb{R}^d$ and $1 \le p < \infty$, the space $C_c(\Omega)$ is dense in $L^p(\Omega)$. This means that for every $f \in L^p(\Omega)$ and $\delta > 0$ there is $g \in C_c(\Omega)$ with $\|f - g\|_p < \delta$.*

This is Rudin, *Real and Complex Analysis*, Theorem 3.14, applied to Lebesgue measure restricted to Ω. For example, a continuous function can approximate a step in integral norm by changing only a short interval near its jump. Uniform approximation of the jump remains impossible.

Write $|y|$ for the Euclidean length of $y \in \mathbb{R}^d$. Define the translation $\tau_y f(x) = f(x - y)$ on $\mathbb{R}^d$. Change of variables, standing fact M5 (Theorem 4.2), gives $\|\tau_y f\|_p = \|f\|_p$ for every p, including ∞.

Theorem 5.5 (Continuity of translation)**.** *For $f \in L^p(\mathbb{R}^d)$ with $1 \le p < \infty$, $\|\tau_y f - f\|_p \to 0$ as $y \to 0$.*

Proof. For $g \in C_c(\mathbb{R}^d)$, uniform continuity gives $\|\tau_y g - g\|_\infty \to 0$. Here uniform continuity means that for each error tolerance one displacement bound works at every point. Continuity on a compact neighbourhood of the support gives such a bound; outside that neighbourhood the function is zero. For $|y| \le 1$, all differences have support in one fixed bounded set of finite measure. Their L^p norms therefore tend to zero too. Given $\delta > 0$, choose g by M6. Translation invariance gives

$$\|\tau_y f - f\|_p \le 2\,\|f - g\|_p + \|\tau_y g - g\|_p .$$

Take the upper limit as $y \to 0$ and then let $\delta \downarrow 0$. ∎

The restriction $p < \infty$ matters. For $f = \mathbb{1}_{(0,1)}$ and $0 < y < 1$, the difference from its translate has modulus 1 on two intervals of total length $2y$. Thus its pth power integral is $2y$, whereas its essential supremum is 1.

5.4 Convolution and approximate identities

The *convolution* of two functions on $\mathbb{R}^d$ is given, where the integral exists, by

$$(f * g)(x) = \int_{\mathbb{R}^d} f(x - y)\,g(y)\,\mathrm{d}y.$$

Change of variables gives commutativity when this integral is absolutely convergent. Convolution uses Lebesgue measure without a normalising factor.

For indicators of intervals, convolution measures an overlap. The figure shows the intervals in the integration variable and the resulting graph.

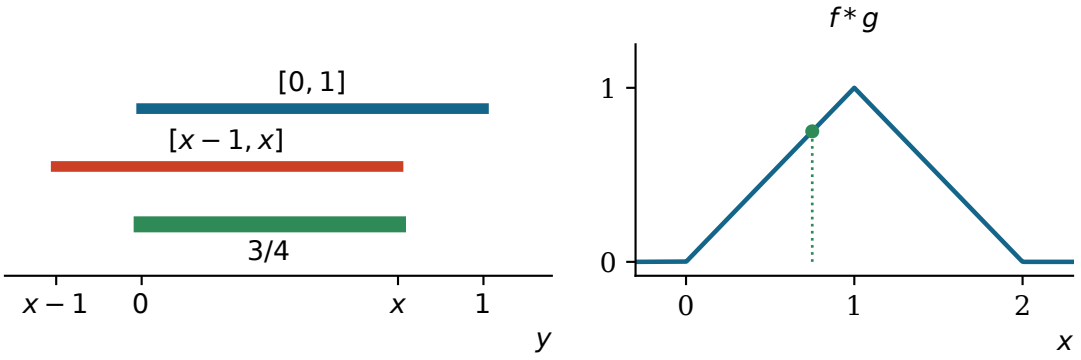


Figure 5.1: The fixed interval is $[0, 1]$ and the translated interval is $[x - 1, x]$, shown at $x = 3/4$. Their common part has length $3/4$. As x varies, this length gives the tent graph.

Example 5.6 (Convolution of two intervals). Take $f = g = \mathbb{1}_{[0,1]}$ on $\mathbb{R}$. For a fixed $x \in \mathbb{R}$, the factor $f(x - y)$ is one exactly when $x - 1 \le y \le x$. The other factor is one on $[0, 1]$. Thus

$$(f * g)(x) = |[0, 1] \cap [x - 1, x]| = \begin{cases} x, & 0 \le x \le 1, \\ 2 - x, & 1 < x \le 2, \\ 0, & x < 0 \text{ or } x > 2. \end{cases}$$

For $0 \le x \le 1$, the overlap runs from zero to x. For $1 < x \le 2$, it runs from $x - 1$ to one. Outside $[0, 2]$ the intervals have no common part of positive length. The output is continuous, although the inputs have jumps. It is not smooth: the slopes change at the corners.

A normalised interval gives a moving average. For $\varepsilon > 0$, set

$$K_\varepsilon(y) = \frac{\mathbb{1}_{[-\varepsilon,\varepsilon]}(y)}{2\varepsilon}, \qquad \int_{\mathbb{R}} K_\varepsilon(y)\, \mathrm{d}y = 1.$$

For a locally integrable function f, meaning one integrable on every bounded interval, commutativity and the substitution $u = x - y$ give

$$(K_\varepsilon * f)(x) = \frac{1}{2\varepsilon} \int_{x-\varepsilon}^{x+\varepsilon} f(u)\, \mathrm{d}u.$$

The value is the average over a window centred at x. Its width shrinks with ε. This interval kernel is not smooth. The smooth kernels used for smoothing are defined below.

The product-measure part of M4 (Theorem 4.1) makes the coordinate lifts $f(x)$ and $g(y)$ measurable on $\mathbb{R}^{2d}$: their sublevel sets are products of measurable sets with $\mathbb{R}^d$. For complex functions, apply this to real and imaginary parts. The invertible linear substitution $(x, y) \mapsto (x - y, y)$ has determinant 1. By M5 (Theorem 4.2), it preserves Lebesgue measurability, including null-set modifications. Thus $f(x - y)g(y)$ is jointly measurable.

This applies also to the absolute values and differences used below. Stein–Shakarchi, *Real Analysis*, Corollary 3.7 and Proposition 3.9, give these measurable-lift consequences of the standing measure and product facts.

Theorem 5.7 (Young's inequality). *If $f \in L^1(\mathbb{R}^d)$ and $g \in L^p(\mathbb{R}^d)$, $1 \le p \le \infty$, then $f * g$ exists absolutely a.e. and $\|f * g\|_p \le \|f\|_1 \|g\|_p$.*

Proof. First apply the nonnegative change of variable $y \mapsto x - y$. It equates the two absolute convolution integrals, even before finiteness is known. Put $A = \|f\|_1$ and suppose $A > 0$. For finite p, Hölder in y with the weight $|f(y)|$ gives

$$\left(\int |f(y)|\, |g(x-y)| \, \mathrm{d}y \right)^p \le A^{p-1} \int |f(y)|\, |g(x-y)|^p \, \mathrm{d}y.$$

For $p = 1$ this is equality. One can first truncate the integrands to apply Hölder, then use monotone convergence (Theorem 3.2). Tonelli, standing fact M4 (Theorem 4.1), and translation invariance show that the integral of the right side in x is $A^p \|g\|_p^p$. It is finite, so the convolution is absolutely convergent a.e., and the bound follows. If $p = \infty$, choose a bounded representative of g. The integral is bounded by $A \|g\|_\infty$. Tonelli on bounded x-sets shows that changes on null sets do not alter the resulting class. If $A = 0$, the assertion is immediate. ∎

An *approximate identity* is a family $K_\varepsilon \in L^1(\mathbb{R}^d)$, for $\varepsilon > 0$, satisfying

$$\int K_\varepsilon = 1, \qquad \sup_{\varepsilon > 0} \|K_\varepsilon\|_1 = M < \infty.$$

For each $\delta > 0$, it must also satisfy the tail condition

$$\int_{|y| > \delta} |K_\varepsilon(y)| \, \mathrm{d}y \longrightarrow 0 \quad \text{as } \varepsilon \downarrow 0.$$

The kernels may be complex and need not be nonnegative.

Theorem 5.8 (Approximate identities). *For $1 \le p < \infty$ and $f \in L^p(\mathbb{R}^d)$, $K_\varepsilon * f \to f$ in L^p. For bounded uniformly continuous f, convergence is uniform.*

Proof. The identity $\int K_\varepsilon = 1$ expresses the error as an integral of $K_\varepsilon(y)(f(x-y) - f(x))$. The map $y \mapsto \|\tau_y f - f\|_p$ is continuous, hence measurable. Indeed, the difference of its values at y and z has modulus at most $\|\tau_{y-z} f - f\|_p$, which tends to zero as $y - z \to 0$ by Theorem 5.5. The weighted Hölder argument above and Tonelli give

$$\|K_\varepsilon * f - f\|_p^p \le M^{p-1} \int |K_\varepsilon(y)| \, \|\tau_y f - f\|_p^p \, \mathrm{d}y.$$

Given $a > 0$, translation continuity makes the norm in the integrand less than a when $|y| < \delta$, for a suitable $\delta > 0$. The integral on this ball is bounded by Ma^p. On its complement use $\|\tau_y f - f\|_p \le 2 \|f\|_p$. The tail condition makes that part tend to zero. The upper limit of the error to the power p is at most $M^p a^p$. Let $a \downarrow 0$. For bounded uniformly continuous f, use the same split directly in the error integral, with $\|\tau_y f - f\|_\infty$ in place of its pth power. ∎

5.5 Mollifiers inside an arbitrary open set

The space $C_c^\infty(\Omega)$ consists of smooth functions whose support is a compact subset of the open set Ω. Write $\mathbb{N}_0 = \mathbb{N} \cup \{0\}$. For a *multi-index* $\alpha = (\alpha_1, \dots, \alpha_d) \in \mathbb{N}_0^d$, use the notation

$$|\alpha| = \sum_{j=1}^{d} \alpha_j, \qquad x^\alpha = \prod_{j=1}^{d} x_j^{\alpha_j}, \qquad \partial^\alpha = \prod_{j=1}^{d} \partial_{x_j}^{\alpha_j}.$$

A *mollifier* is a nonnegative $j \in C_c^\infty(\mathbb{R}^d)$ with $\operatorname{supp} j \subset B(0,1)$ and $\int j = 1$. Such a function exists: normalise $\exp(-1/(1-4\,|x|^2))$ on $|x| < 1/2$ and set it to zero elsewhere. Every interior derivative is an exponential times a rational function of the coordinates and $1 - 4\,|x|^2$. Every such derivative tends to zero at the boundary, since $s^{-N} e^{-1/s} \to 0$ as $s \downarrow 0$ for every integer $N \geq 0$. Extend each derivative by zero across the sphere. These extensions are continuous. At a boundary point, the mean value theorem on a coordinate segment bounds a difference quotient by the next interior derivative, which tends to zero. A segment lying outside the ball has zero quotient. Induction on the derivative order therefore proves that the zero extension is smooth, as for the flat function in § 1.4. Define $j_\varepsilon(x) = \varepsilon^{-d} j(x/\varepsilon)$. Change of variables gives $\int j_\varepsilon = \|j_\varepsilon\|_1 = 1$. Its support lies in $B(0,\varepsilon)$, so these kernels form an approximate identity.

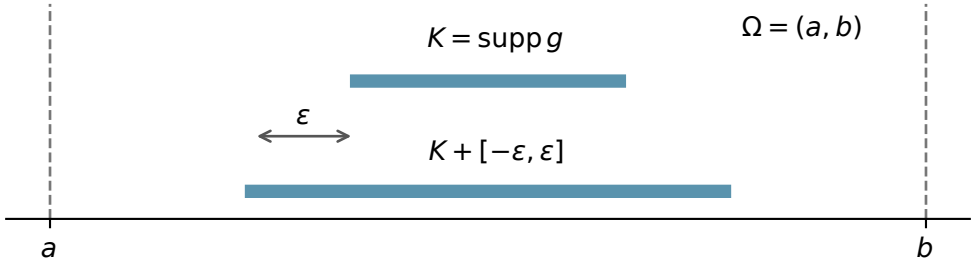
Theorem 5.9 (Smooth density). *For any open $\Omega \subset \mathbb{R}^d$ and $1 \leq p < \infty$, the space $C_c^\infty(\Omega)$ is dense in $L^p(\Omega)$.*

Proof. Given $f \in L^p(\Omega)$ and $a > 0$, M6 gives $g \in C_c(\Omega)$ with $\|f - g\|_p < a/2$. If $g = 0$ there is nothing to smooth. Otherwise its support K is compact in Ω. There is $r > 0$ such that the closed r-neighbourhood of K lies in Ω: cover K by finitely many balls whose doubled closed balls lie in Ω, and take a sufficiently small minimum radius. Extend g by zero to $\mathbb{R}^d$. The extension is continuous and compactly supported. For $0 < \varepsilon < r$, put $g_\varepsilon = j_\varepsilon * g$. Its support lies in $K + \overline{B(0,\varepsilon)}$, a compact subset of Ω. Differentiation under the integral gives

$$\partial^\alpha g_\varepsilon(x) = \int (\partial^\alpha j_\varepsilon)(x - y) g(y)\, \mathrm{d}y.$$

For each derivative, the bounded derivatives of j_ε times $|g| \in L^1$ dominate the difference quotients locally in x. The parameter-integral theorem (Theorem 4.4) therefore applies repeatedly. Thus $g_\varepsilon \in C_c^\infty(\Omega)$. The approximate identity theorem makes $\|g_\varepsilon - g\|_p < a/2$ for small ε. The triangle inequality finishes the proof. ∎

The support condition is needed even for an interval. The diagram shows how smoothing enlarges a compact support while leaving room at the boundary.



The lower interval bounds the support after convolution. Its ends move by at most ε. Choose ε smaller than the distance of the original support from the complement of Ω.

The *Schwartz space* $\mathcal{S}(\mathbb{R}^d)$ consists of smooth functions for which every $x^\alpha \partial^\beta f$ is bounded, for multi-indices $\alpha, \beta \in \mathbb{N}_0^d$. Every compactly supported smooth function belongs to this space. Hence $\mathcal{S}$ is dense in every finite $L^p(\mathbb{R}^d)$. Lecture 7 studies its derivatives and Fourier transform. The density theorem applies to bounded intervals and disconnected open sets as well as to $\mathbb{R}^d$. It supplies QM-II's smooth approximation assumption.

5.6 The Hilbert space of square-integrable functions

An inner product in this course is antilinear in its first argument. On $L^2(E)$, define

$$\langle f, g \rangle = \int_E \overline{f(x)} g(x)\, \mathrm{d}x.$$

Hölder with $p = 2$ proves that the integral exists and $|\langle f, g\rangle| \le \|f\|_2 \, \|g\|_2$. Linearity in the second argument, conjugate symmetry and positivity follow from the integral. Positive definiteness holds for classes, since $\langle f, f\rangle = 0$ implies $f = 0$ a.e. A *Hilbert space* is an inner product space complete in the induced norm.

Theorem 5.10 (L^2 completeness)**.** *The space $L^2(E)$ with this inner product is a Hilbert space.*

Proof. The induced norm is $(\langle f, f\rangle)^{1/2} = \|f\|_2$. Completeness is Theorem 5.3 with $p = 2$. This discharges the completeness assumption cited in Linear Algebra's introduction to Hilbert spaces. ∎

For example, on $(0, 1)$ the wavefunction $\psi(x) = \sqrt{3}x$ has norm one, since $\int_0^1 3x^2 \, \mathrm{d}x = 1$. Its value at an individual point has no meaning as an L^2 observable. The probability of a measurable set $A \subset (0, 1)$ is $\int_A |\psi|^2$, which is unchanged by null-set modifications.

Summary.

- The quotient spaces $L^p(E)$ are Banach spaces for $1 \le p \le \infty$. A norm-convergent sequence has an a.e.-convergent subsequence.
- Translation is continuous in finite $L^p(\mathbb{R}^d)$. Young's inequality defines convolution by an L^1 function on every L^p.
- Approximate identities converge in finite L^p, and uniformly on bounded uniformly continuous functions.
- For every open Ω, compactly supported smooth functions are dense in $L^p(\Omega)$ when $p < \infty$. The space L^2 is a Hilbert space.

Lecture 6 applies approximation to Fourier series.

References

E. M. Stein and R. Shakarchi, *Functional Analysis*, Princeton, 2011, Chapter 1, §§1–2. W. Rudin, *Real and Complex Analysis*, 3rd ed., 1987, Theorems 3.11–3.14. E. M. Stein and R. Shakarchi, *Real Analysis*, 2005, Chapter 2, Corollary 3.7 and Proposition 3.9. Further reading: R. S. Strichartz, *The Way of Analysis*, revised edition, 2000, §7.5.2 and §14.4.